Local risk bounds for statistical aggregation

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Abstract

We study statistical aggregation under the squared loss: given a reference (possibly non-convex) family of predictors, the goal is to output a single predictor whose risk is close to the best achievable within the family. Allowing improper learning via mixtures convexifies the problem and enables fast excess-risk rates. Building on recent advances, we present risk bounds for two central aggregation schemes, exponential weights (EW) and Q-aggregation, showing how classical bounds based on *global* entropic complexity can be refined through a *local* complexity measure that concentrates the prior around good predictors. In particular, we report a localized bound in expectation for EW under Gaussian noise and a deviation-optimal high-probability bound for Q-aggregation under sub-Gaussian noise, and we outline the proof mechanism for Q-aggregation based on strong convexity and convex duality, highlighting why the same strategy fails for EW. Finally, we discuss practical computations: we approximate EW using grid methods in low dimension and Langevin-type sampling in higher dimension, and we comment on the moment-based characterization of Q-aggregation and its implications for implementability beyond structured settings.

1 Introduction

A recurring goal in supervised learning is the following: given a reference family of predictors $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ and a loss ℓ (here, the squared loss), construct a single predictor $\widehat{f}$ whose risk is nearly as small as the best risk achievable within the class, i.e. we want to find $\widehat{f}$ which minimizes:

$$R(\widehat{f}) - \inf_{\theta\in\Theta} R(f_\theta) \tag{1}$$

where $R(f) = \mathbb{E}\big[\ell(Y, f(X))\big]$ denotes the (population) risk and $R_n(f) = \frac{1}{n}\sum_{i=1}^n \ell(Y_i, f(X_i))$ the empirical risk.
A natural baseline is *model selection* (proper learning): choose a single candidate that minimizes the empirical risk,

$$\widehat{\theta} \in \arg\min_{\theta\in\Theta} R_n(f_\theta), \qquad \widehat{f} = f_{\widehat{\theta}}. \tag{2}$$

The motivation for aggregation becomes clear already in the important special case of a *finite dictionary* $|\Theta| = M$. In the generic setting emphasized by the paper, the family $\mathcal{F}$ is not assumed to enjoy helpful structural properties; in particular, it may be *non-convex* and the target need not be well aligned with the class. In such regimes, procedures that must output one of the M candidates (deterministic or randomized selection rules) can exhibit inherently slower worst-case behavior: the excess risk cannot generally improve upon the rate

$$R(\widehat{f}) - \min_{\theta\in\Theta} R(f_\theta) \gtrsim \sqrt{\frac{\log M}{n}}, \tag{3}$$

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reflecting the fact that non-convexity obstructs the usual acceleration one obtains from the curvature of the squared loss [1].

Aggregation addresses this limitation by allowing *improper* predictors obtained by mixing models. Instead of selecting a single θ, one selects a probability distribution ρ over Θ and predicts with the mixture

$$f_\rho(x) \;=\; \int_\Theta f_\theta(x)\,\rho(d\theta), \tag{4}$$

so that the search space becomes the simplex (or, more generally, $\mathcal{P}(\Theta)$), which is *convex*. This "convexification" is crucial: in a convex set, the local curvature of the squared loss can be exploited to control fluctuations and obtain fast rates. In the finite dictionary case, this enables aggregation procedures to achieve the optimal order as shown in [2], which is:

$$R(\widehat{f}) - \min_{\theta\in\Theta} R(f_\theta) \lesssim \frac{\log M}{n}. \tag{5}$$

Building on this motivation, the paper "Local Risk Bounds for Statistical Aggregation" refines classical statistical-aggregation guarantees by showing that global complexity terms can often be replaced by *local* entropic complexity measures, yielding sharper risk bounds (in expectation and with high probability) for prominent aggregation schemes such as exponential weights and Q-aggregation.

2 Exponential weights and Q-aggregation

2.0.1 Exponential Weights (EW)

Definition. Let π be a prior distribution on Θ and $\beta > 0$ an inverse temperature parameter. The *exponential weights* aggregation method defines

$$\widehat{\rho}_{\mathrm{EW}} \;\in\; \arg\min_\rho \left\{ \int_\Theta R_n(\theta)\,\rho(d\theta) \;+\; \frac{1}{\beta}\,\mathrm{KL}(\rho\|\pi) \right\}, \tag{6}$$

and outputs the aggregated predictor $\widehat{f}_{\mathrm{EW}} := f_{\widehat{\rho}_{\mathrm{EW}}}.$, $\widehat{\rho}_{\mathrm{EW}}$ has the Gibbs form

$$\widehat{\rho}_{\mathrm{EW}}(d\theta) \;\propto\; \exp\{-\beta R_n(\theta)\}\,\pi(d\theta). \tag{7}$$

EW assigns a weight to each predictor according to its empirical risk: predictors with smaller $R_n(\theta)$ receive exponentially larger weight. The KL term penalizes deviations from the prior π, which stabilizes the procedure and prevents over-concentration unless the data strongly supports it. The parameter β controls how selective the weighting is: large β concentrates the mass on the empirically best predictors (close to model selection), while small β spreads the weights more broadly (more averaging and regularization).

2.0.2 Q-Aggregation

Definition. With π a prior on Θ, $\beta > 0$, *Q-aggregation* is defined by

$$\widehat{\rho}_{\mathrm{Q}} \;\in\; \arg\min_\rho \left\{ \frac{1}{2}\int_\Theta R_n(\theta)\,\rho(d\theta) \;+\; \frac{1}{2}R_n(\rho) \;+\; \frac{1}{\beta}\,\mathrm{KL}(\rho\|\pi) \right\}, \tag{8}$$

and outputs $\widehat{f}_{\mathrm{Q}} := f_{\widehat{\rho}_{\mathrm{Q}}}$.

Q-aggregation differs from EW by adding the term $R_n(\rho)$, i.e., the empirical risk of the aggregated predictor itself. This directly encourages the final mixture to perform well, not only the individual predictors receiving weight. Under squared loss, this additional term introduces useful strong conevexity: it favors mixtures that benefit from averaging (typically reducing instability/variance) while the linear part $\int R_n(\theta)\rho(d\theta)$ still pushes toward strong individual models (a soft model-selection effect). As in EW, the KL penalty keeps the solution close to the prior unless the data provides evidence to move away from it.

2.1 Bounds on the risk for EW and Q-agg

2.1.1 Existing literature review

The existing literature already provides optimal risk bounds for the exponential weights (EW) estimator in expectation, as well as deviation bounds for the Q-aggregation estimator. Here, optimality refers to the fact that these results imply an excess risk of order $\mathcal{O}(1/n)$. Such bounds can be found in [3] and [4].

However, these results are derived in a unified framework that applies equally to the statistical and the sequential settings. This is unsatisfactory, since the sequential setting is inherently more challenging. Indeed, in the sequential framework, observations are revealed one at a time, and the analysis must account for the worst-case ordering of the data. In contrast, in the statistical setting, the entire sample is observed at once, which allows for more accurate inference.

This motivates the search for bounds that are tailored to the statistical setting and potentially sharper than those obtained by directly transferring results from the sequential framework.

2.1.2 Global and local complexities

It has been proven in the original paper that the following inequalities hold for c ≥ 8:

$$\frac{1}{n+1}\sum_{i=0}^{n}\mathbb{E}\Big[R\big(\hat{f}_i^{\mathrm{iew}}\big)\Big] \;\leq\; \inf_{\gamma\in\mathcal{P}(\Theta)}\left\{\mathbb{E}_{\theta\sim\gamma}[\,R(\theta)\,] \;+\; \frac{c}{n+1}\,\mathrm{KL}(\gamma\,\|\,\pi)\right\} := C_{glob}(\frac{n+1}{c},\pi)$$

$$\mathbb{E}\Big[R\big(\hat{f}_n^{\mathrm{pm}}\big)\Big] \;\leq\; C_{\mathrm{glob}}\left(\frac{n+1}{c},\pi\right).$$

where: $\hat{f}_n^{\mathrm{pm}} = \frac{1}{n+1}\sum_{i=0}^{n}\hat{f}_i^{\mathrm{iew}}$

The quantity on the right of both inequalities is called the global complexity. The first inequality shows that it bounds the sequential case, and the second inequality shows that it bounds the statistical case. By looking at its definition, we see that it is defined by a minimization problem which aims at finding the probability measure that achieves the best tradeoff between having a small average risk (first term of the sum) and staying close enough to the prior (second term of the sum).

The refinement proposed by the authors is the following quantity, which they call local complexity:

$$C_{\mathrm{loc}}(\beta,\pi) = \frac{\int R(\theta)\,e^{-\beta R(\theta)}\,\pi(d\theta)}{\int e^{-\beta R(\theta)}\,\pi(d\theta)} = \mathbb{E}_{\theta\sim\pi_{-\beta R}}[R(\theta)]$$

where $\pi_h(d\theta) \;=\; \frac{e^{h(\theta)}\,\pi(d\theta)}{\int_\Theta e^{h(\vartheta)}\,\pi(d\vartheta)}$

The measure $\pi_{-\beta R}$ reshapes the prior π around good models only. Indeed, the bad models are given exponentially smaller weights than the good models. However, it can't create mass that does not exist: if the prior is too bad and does not put any mass on some of the good models, the measure $\pi_{-\beta R}$ won't put any mass on these models either.

Relationships between complexities We can actually rewrite the global complexity as:

$$C_{\mathrm{glob}}(\beta,\pi) = \mathbb{E}_{\theta\sim\pi_{-\beta R}}[R(\theta)] + \frac{1}{\beta}\,\mathrm{KL}(\pi_{-\beta R}\,\|\,\pi) = C_{\mathrm{loc}}(\beta,\pi) + \frac{1}{\beta}\,\mathrm{KL}(\pi_{-\beta R}\,\|\,\pi)\,.$$

As a consequence, the local complexity is always smaller than the global one. Both are equal if and only if the prior is actually exactly the Gibbs posterior. This is a key result, since the quantity used to refine the bounds from the sequential setting is exactly the local complexity.

2.1.3 Risk bounds achieved by the authors

Theorem 1 (Localized bound for EW (expectation))**.** *Assume the fixed-design regression model $Y_i = f^\star(X_i) + Z_i$ with i.i.d. centered Gaussian noise $Z_i \sim \mathcal{N}(0, \sigma^2)$. Consider $\widehat{f}_n^{\mathrm{EW}}(\pi, \beta)$ with $\beta = n/(8\sigma^2)$. Then*

$$\mathbb{E}\big[R(\widehat{f}_n^{\mathrm{EW}})\big] \;\le\; \mathcal{C}_{\mathrm{loc}}\Big(\frac{n}{16\sigma^2}, \pi\Big)\,.$$

Theorem 2 (Deviation-optimal bound for Q-aggregation)**.** *Assume the fixed-design regression model $Y_i = f^\star(X_i) + Z_i$ where $(Z_i)_{i=1}^n$ are i.i.d., centered and σ^2-sub-Gaussian. Let $\widehat{f}_n^{\mathrm{Q}}(\pi, \beta)$ be the Q-aggregation estimator with $\beta = n/(12\sigma^2)$. Then for any $\delta \in (0,1)$, with probability at least $1-\delta$,*

$$\big\|\widehat{f}_n^{\mathrm{Q}} - f^\star\big\|_n^2 \;\le\; \inf_{\gamma' \in \mathcal{P}(\Theta)} \left\{ \big\langle \gamma',\, \|f - f^\star\|_n^2 \big\rangle + \frac{3}{2\beta}\,\mathrm{KL}\Big(\gamma' \,\Big\|\, \pi_{-\frac{\beta}{3}R}\Big) \right\} + \frac{3\log(1/\delta)}{2\beta}. \tag{9}$$

In particular, by choosing $\gamma' = \pi_{-\frac{\beta}{3}R}$ in the infimum,

$$R\Big(\widehat{f}_n^{\mathrm{Q}}\Big) \;\le\; \mathcal{C}_{\mathrm{loc}}\left(\frac{\beta}{3}, \pi\right) + \frac{3\log(1/\delta)}{2\beta}. \tag{10}$$

Proof sketch of Theorem 2. Write $\widehat{\rho} := \widehat{\rho}_{\mathrm{Q}}$ and $\widehat{f} := f_{\widehat{\rho}}$. Fix a comparison distribution $\gamma \in \mathcal{P}(\Theta)$ and denote $f_\gamma := \int f_\theta\, \gamma(d\theta)$. We study the gap between the empirical error of $\widehat{f}$ and a localized benchmark:

$$\Delta(\widehat{\rho}, \gamma) := \big\|\widehat{f} - f^\star\big\|_n^2 - \inf_{\gamma' \in \mathcal{P}(\Theta)} \left\{ \big\langle \gamma',\, \|f - f^\star\|_n^2 \big\rangle + \frac{3}{2\beta}\,\mathrm{KL}\Big(\gamma' \,\Big\|\, \pi_{-\frac{\beta}{3}R}\Big) \right\}.$$

Step 1: Decomposition. Using $Y = f^\star + Z$ (componentwise), one can expand the square and isolate two contributions: an *empirical gap* and a *noise process* (plus KL/variance terms that will be handled by localization):

$$\Delta(\widehat{\rho}, \gamma) = \underbrace{R_n(\widehat{\rho}) - R_n(\gamma)}_{\text{(I) empirical gap}} + \underbrace{\frac{2}{n}\big\langle \widehat{f} - f_\gamma,\, Z\big\rangle_n}_{\text{(II) noise process}} + \cdots$$

where $\langle u, v\rangle_n := \sum_{i=1}^n u_i v_i$ and $\|\cdot\|_n^2 := \frac{1}{n}\langle \cdot, \cdot\rangle_n$.

Step 2: Control of the empirical gap. Because the squared loss is strongly convex and $\widehat{\rho}$ minimizes the Q-aggregation objective, one gets a one-sided control of $R_n(\widehat{\rho}) - R_n(\gamma)$ featuring a negative quadratic term:

$$R_n(\widehat{\rho}) - R_n(\gamma) \;\le\; -\frac{1}{2}\Big\langle \widehat{\rho},\, \|f - f_\gamma\|_n^2 \Big\rangle \;+\; \cdots$$

After standard algebraic rearrangements (this is where the localization lemma is used to merge/cancel the remaining KL and variance terms), this yields a deterministic bound of the form

$$\frac{2\beta}{3}\Delta(\widehat{\rho}, \gamma) \;\le\; \langle \widehat{\rho},\, h\rangle - \mathrm{KL}\Big(\widehat{\rho} \,\Big\|\, \pi_{-\frac{\beta}{3}R}\Big)\,, \tag{11}$$

for a suitable function h (depending on Z) given by

$$h(\theta) = -\frac{\beta}{6}\,\|f_\theta - f_\gamma\|_n^2 + \frac{2\beta}{n}\,\langle f_\theta - f_\gamma,\, Z\rangle_n.$$

Step 3: Convex duality (change of measure). Applying the convex-conjugate identity

$$\sup_{\rho \in \mathcal{P}(\Theta)} \big\{ \langle \rho, h\rangle - \mathrm{KL}(\rho \| \pi) \big\} = \log\Big(\mathbb{E}_{\theta\sim\pi}[e^{h(\theta)}]\Big)$$

to (11) (with $\pi = \pi_{-\frac{\beta}{3}R}$) gives

$$\frac{2\beta}{3}\,\Delta(\widehat{\rho}, \gamma) \;\le\; U \;:=\; \log\Big(\mathbb{E}_{\theta\sim\pi_{-\frac{\beta}{3}R}} \exp\Big(-\frac{\beta}{6}\|f_\theta - f_\gamma\|_n^2 + \frac{2\beta}{n}\langle f_\theta - f_\gamma,\, Z\rangle_n\Big)\Big).$$

Step 4: Sub-Gaussian control of the noise process. By independence and σ^2-sub-Gaussianity of (Z_i), the moment generating function of $\langle f_\theta - f_\gamma,\ Z\rangle_n$ is controlled, and for $\beta \le n/(12\sigma^2)$ one obtains

$$\mathbb{E}_Z\big[e^U\big] \ \le\ 1.$$

Therefore, by Markov's inequality, for any $\delta \in (0,1)$,

$$\mathbb{P}\Big(U \ge \log(1/\delta)\Big) \ \le\ \delta, \qquad \text{i.e.} \qquad U \le \log(1/\delta) \ \text{ with prob. } \ge 1-\delta.$$

Step 5: Conclusion. On the event $\{U \le \log(1/\delta)\}$, we have

$$\Delta(\widehat{\rho},\gamma) \ \le\ \frac{3}{2\beta}\log(1/\delta).$$

Unpacking the definition of $\Delta(\widehat{\rho},\gamma)$ yields (9). Finally, (10) follows by choosing $\gamma' = \pi_{-\frac{\beta}{3}R}$ in the infimum. □

Why the same strategy fails for EW. It is tempting to try to replicate the previous argument for the EW aggregate, since EW also arises from a variational formulation with a KL penalty. However, the key step in the Q-aggregation proof is to apply a strong-convexity inequality to the *Q-objective*. Because $\widehat{\rho}_{\mathrm{Q}}$ is the minimizer of this same strongly convex objective, the associated first-order (gradient) term vanishes at $\widehat{\rho}_{\mathrm{Q}}$, and one is left with a *negative quadratic term*. This quadratic term is precisely what allows the empirical gap to compensate the noise process, and it is what makes it possible to take a supremum over all measures ρ and then invoke the convex duality formula.

For EW, this mechanism breaks down. EW is *not* optimal for the strongly convex Q-objective, so if we apply the same strong-convexity inequality at $\widehat{\rho}_{\mathrm{EW}}$, the gradient term does *not* vanish. As a consequence, we cannot derive the clean inequality that isolates a negative quadratic term capable of cancelling the noise process through the empirical gap. Since this cancellation is missing, we can no longer control the supremum over ρ, and therefore we cannot apply the convex duality identity in the same way.

3 Extensions and Experiments

3.1 Larger class of noise

We can actually extend the results on the EW estimator for a larger class of noise. For a precise description of the class considered, we refer to [5]. This class includes gaussian noise but also centered bernoulli, centered binomial, and laplace. However, the bound is only controlled with the global complexity. If we wished to extend the results to the local complexity, we would need to completely chnage the proof of the first theorem in the original paper, as it heavily relies on the assumption that the noise is Gaussian.
To our knowledge, there is no existing work aiming at extending the results of Theorem 2 (related to the Q-aggregation) to a larger class of noise.

3.2 Other losses

The original paper only deals with the squared loss. We adressed the question to wether we can extend the results to another loss. As the squared loss is convex, maybe convexity is enough to get optimal bounds on the excess risk rate. However, this result, shown in [6], displays that it is actually not the case:
Let $l(f(x),y) = \max(0, 1 - f(x)y)$. One easily checks that l is convex in f(x). For all $M \in \mathbb{N}$, $\exists f_1, \dots, f_M, P$ such that $(X,Y) \sim P$, and such that for any aggregation procedure:

$$\mathbb{E}[R(\hat{f}_n)] - \min_{f\in F} R(f) \sim \left(\frac{\log M}{n}\right)^{\alpha}, \quad \alpha < 1$$

which is a suboptimal rate.

Then, we investigated if strong convexity was enough. Under some other assumptions, it is indeed the case, as it has been shown in [7], Theorem A.

However, it is hard to construct such a loss in regression which is not the squared loss, even though adding a convex term to the squared loss would indeed yield a loss that satisfies the hypothesis needed.
In the classification setting, one easily checks that the logistic loss satisfies the hypothesis.

3.3 Computation of the Exponential Weight estimator in practice

The exponential weight estimator actually admits an explicit Gibbs-form expression, which defines the following density:

$$\frac{\mathrm{d}\rho_n^{\mathrm{ew}}}{\mathrm{d}\pi}(\theta) = \frac{\exp\bigl(-\beta R_n(\theta)\bigr)}{\int \exp\bigl(-\beta R_n(\vartheta)\bigr)\,\pi(\mathrm{d}\vartheta)}.$$

which then yields:

$$\hat{f}^{ew}(x) = \frac{\int f_\vartheta(x) e^{-\beta R_n(\vartheta)} \pi(d\vartheta)}{\int e^{-\beta R_n(\vartheta)} \pi(d\vartheta)}.$$

However, the integrals on the numerator and denominator do not admit a closed form in the general case. Hence, we would like to approximate them using a grid technique.

Grid-based approximation. Let π be a probability measure on $\mathbb{R}^d$ admitting a density $p(\theta)$ with respect to the Lebesgue measure, i.e., $\pi(d\theta) = p(\theta)\, d\theta$. The integrals defining the exponential weights estimator are approximated by discretizing the parameter space over a finite grid $\{\theta_1, \ldots, \theta_M\}$. For simplicity, we consider a uniform grid with constant volume element $\Delta\theta$. This yields:

$$s_j = \exp\bigl(-\beta R_n(\theta_j)\bigr)\, p(\theta_j)\, \Delta\theta, \qquad Z = \sum_{j=1}^{M} s_j, \qquad w_j = \frac{s_j}{Z}.$$

Here $\Delta\theta$ corresponds to the volume element associated with the grid discretization. This yields the following approximation of the aggregated predictor:

$$\hat{f}^{\mathrm{ew}}(x) \approx \sum_{j=1}^{M} w_j f_{\theta_j}(x).$$

While this approach is straightforward and effective in low-dimensional settings, it quickly becomes computationally prohibitive as the dimension d increases, due to the exponential growth of $M = k^d$ (curse of dimensionality).

Solution based on Langevin dynamics We propose a solution based on Langevin dynamics. In [8], the authors already proved convergence results for this method in the Exponential Weigtht aggregation context. However, all of these results use a certain type of prior, which does not include basic priors such as Gaussian, uniform, or Laplace. Hence, we adapted the classical convergence results of Langevin dynamics found in the literature to the specific EW estimation context.

Consider the Langevin stochastic differential equation

$$\mathrm{d}\theta_t = -\nabla U(\theta_t)\,\mathrm{d}t + \sqrt{2}\,\mathrm{d}W_t,$$

where $U : \mathbb{R}^d \to \mathbb{R}$ is a potential function and $(W_t)_{t\geq 0}$ is a standard d-dimensional Brownian motion. It is well known (see, e.g. [9], theorems 1.2 and 1.6) that the following assumptions are sufficient to guarantee the existence of a unique invariant probability measure and convergence in law of the Langevin dynamics towards stationary distribution.

A1(m) U is m-convex, i.e. there exists $m \geq 0$ such that for all $x, y \in \mathbb{R}^d$ and $t \in [0,1]$,

$$U(tx + (1-t)y) \leq tU(x) + (1-t)U(y) - \frac{m}{2}t(1-t)\|x-y\|^2.$$

Equivalently, U is twice differentiable and

$$\nabla^2 U(\theta) \succeq mI_d \quad \forall \theta \in \mathbb{R}^d.$$

(which means that $\nabla^2 U(\theta) - mI_d$ is positive semi-definite).

A2 U is continuously differentiable and has L-Lipschitz gradient, i.e. there exists $L > 0$ such that

$$\|\nabla U(x) - \nabla U(y)\| \leq L\|x-y\| \quad \forall x, y \in \mathbb{R}^d.$$

Under Assumptions **A1**(m) and **A2**, the Langevin diffusion admits a unique invariant distribution with density proportional to $\exp(-U(\theta))$.

Application to the exponential weights posterior. In our setting, the potential function is given by

$$U(\theta) = \beta R_n(\theta) - \log \pi(\theta),$$

where R_n denotes the empirical risk associated with the squared loss and π is the prior distribution on the parameter θ. We then look at priors which are such that $U(\theta)$ satisfy both of the above assumptions.

Assume a linear prediction model and squared loss, so that

$$R_n(\theta) = \frac{1}{n}\sum_{i=1}^{n}(x_i^\top \theta - y_i)^2.$$

A direct computation yields

$$\nabla^2 R_n(\theta) = \frac{2}{n}X^\top X,$$

where $X \in \mathbb{R}^{n \times d}$ is the design matrix. In particular, $\nabla^2 R_n(\theta)$ is positive semidefinite and constant in θ.

Let the prior π be Gaussian, $\pi = \mathcal{N}(0, \Sigma)$, with Σ positive definite. Then

$$-\log \pi(\theta) = \frac{1}{2}\theta^\top \Sigma^{-1}\theta + C,$$

and hence

$$\nabla^2(-\log \pi)(\theta) = \Sigma^{-1}.$$

Combining the two contributions, the Hessian of U is given by

$$\nabla^2 U(\theta) = \beta\frac{2}{n}X^\top X + \Sigma^{-1}.$$

Since Σ^{-1} is positive definite, it follows that

$$\nabla^2 U(\theta) \succeq \lambda_{\min}(\Sigma^{-1})I_d,$$

and therefore Assumption **A1**(m) holds with

$$m = \lambda_{\min}(\Sigma^{-1}) > 0.$$

Moreover, since $\nabla^2 U(\theta)$ is constant in θ, the gradient ∇U is globally Lipschitz with constant

$$L = \beta\frac{2}{n}\|X^\top X\|_{\mathrm{op}} + \|\Sigma^{-1}\|_{\mathrm{op}},$$

so that Assumption **A2** is also satisfied.

As a consequence, in the case of a squared loss and a Gaussian prior, the associated Langevin diffusion converges to a unique stationary distribution, which coincides with the exponential weights (Gibbs) posterior with density proportional to $\exp(-U(\theta))$.

We do not prove it here, but the Uniform prior (if R_n is m-convex) and the Laplace prior also satisfy the assumptions.
Now, for practical implementation, we need to discretize the SDE. We use the Euler-Maruyama scheme to get:

$$\theta_{t+1} = \theta_t - \tau_{t+1}\nabla U(\theta_t) + \sqrt{2\tau_{t+1}}B_{t+1}$$

We would still like to get some guarantee of convergence for the discrete scheme towards the stationary distribution. When a constant step size $\tau_t = \tau$ is used, the resulting Markov chain admits a stationary distribution π_τ, which generally differs from the target distribution π, but converges to π as $\tau \to 0$ ([10]). In contrast, when the step sizes satisfy $\tau_t \to 0$ and $\sum_{t=1}^{\infty} \tau_t = \infty$, the algorithm allows for a direct approximation of π without asymptotic bias, as proven in [11] and [12]. Furthermore, non-asymptotic convergence guarantees have been established in total variation distance, for example in [13] and [14].

This allows us to use a constant step size in practice, provided it is small enough, at the price of a controlled discretization bias.

Now, we would like to get that the Markov chain resulting from the discretization is ergodic in the case of constant step size, in order to use the ergodic theorem. In Theorems 5.4 and 5.5 of [15], we get the result for U satisfying both of the above hyppthesis (to be precise, in the paper it is for potentials satisfying **A1**(m) and such that the growth is at most quadratic, but this is implied by **A2**). Hence we indeed have:

$$\frac{1}{T}\sum_{t=0}^{T} f_{\theta_t}(x) \ \to \mathbb{E}_{\theta\sim\hat{\rho}_{n_\tau}^{ew}}[f_\theta(x)]$$

where $\hat{\rho}_{n_\tau}^{ew}$ is the stationnary distribution of the chain, which converges to $\hat{\rho}_n^{ew}$ in total variation as τ goes to 0.

In practice, it is often the case that a "burn-in" period is used, to avoid the first samples that are too far from the stationaru distribution. We then need to rescale in order to get:

$$\frac{1}{T-t_0}\sum_{t=t_0}^{T} f_{\theta_t}(x) \ \to \mathbb{E}_{\theta\sim\hat{\rho}_{n_\tau}^{ew}}[f_\theta(x)]$$

Numerical results: We will implement the Langevin dynamics with constant step size, which converges to a distribution close to the stationary one for a small enough step size. Corollary 1 in [14] gives us the learning rate and number of iterations we need to choose to get a given precision ϵ, in function of ϵ. However, because of the nature of the proof, these theoretical choices are highly pessimistic. Hence, we choose a higher step size and a smaller number of iterations, which will still yield good results.

Note: All of the details of the implementations will be given in the appendix.

We first evaluate the method when we have a closed form for the EW estimator (which is the case for linear functions and gaussian prior). We use the euclidean norm on R^d to measure the distance between the parameter θ of the closed form and the one we get from the approximation, and this quantity is plotted in function of the number of steps after burn-in. We see that the algorithm with fixed step size gives us a good approximation after enough iterations.

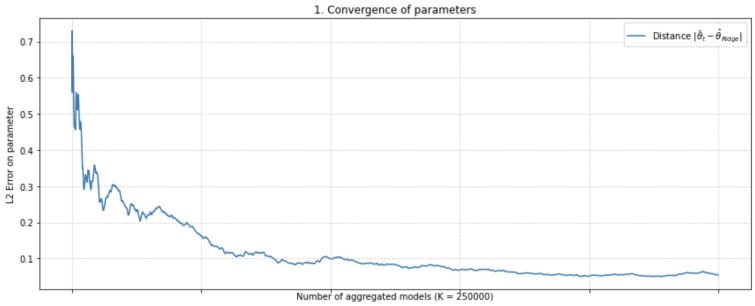


Figure 1: Convergence of the ULA towards the true solution

Then, we use the algorithm to compute the EW estimator in the case of Laplace prior and linear functions, and try to recover the first theorem of the paper, which bounds the expected risk of the EW estimator with the local complexity.

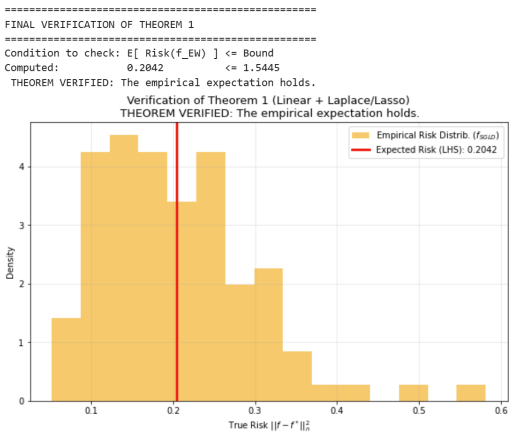


Figure 2: Histogram representing the risk of the samples obtained via ULA

As described above the figure, the theorem is verified: we indeed find that the local complexity bounds the expected risk of the EW estimator. The bound is not plotted for scales reason: the bound is roughly ten times higher than the expected risk. One might think that this implies a bad convergence of our algorithm. However, the spread between the expected risk and the bound is in the same order as the one from the closed form case: the bound is loose because of the nature of the proof.

3.4 Computation of the Q-aggregation estimator in practice

Practical computation in the finite dictionary setting. In our experiments, we work with a finite collection of candidate predictors $\{f_{\theta_1}, \dots, f_{\theta_M}\}$ evaluated on the design points $(x_i)_{i=1}^n$. We store these evaluations in a matrix $P \in \mathbb{R}^{n \times M}$ defined by $P_{i,j} = f_{\theta_j}(x_i)$, and denote by $Y \in \mathbb{R}^n$ the vector of responses. The aggregation weights are a vector $\rho \in \Delta_M$ on the probability simplex $\Delta_M = \{\rho \in \mathbb{R}_+^M : \sum_{j=1}^M \rho_j = 1\}$, so that the aggregated predictor on the design points is simply

$$u(\rho) = P\rho \in \mathbb{R}^n, \qquad u_i(\rho) = \sum_{j=1}^M \rho_j f_{\theta_j}(x_i).$$

We also precompute the empirical risk of each candidate estimator,

$$\ell_j = \frac{1}{n} \sum_{i=1}^n \big(f_{\theta_j}(x_i) - Y_i\big)^2, \qquad j = 1, \dots, M,$$

and we denote by $\pi \in \Delta_M$ the prior weights (normalized to sum to one).

We then compute the Q-aggregation weights as the solution of the following regularized optimization problem (for squared loss):

$$\min_{\rho \in \Delta_M} \; J(\rho) = \alpha \langle \rho, \ell \rangle + (1-\alpha)\,\frac{1}{2n}\|P\rho - Y\|_2^2 + \frac{1}{\beta}\,\mathrm{KL}(\rho\|\pi),$$

where $\beta > 0$ is the temperature parameter and $\mathrm{KL}(\rho\|\pi) = \sum_{j=1}^{M} \rho_j \log(\rho_j/\pi_j)$. The parameter $\alpha \in [0,1]$ is equal to $\frac{1}{2}$ in the case of the Q-aggregation method.

Since J is the sum of a linear term, a convex quadratic term, and a convex KL regularizer, the problem is convex on Δ_M. We minimize it using projected gradient descent with backtracking line search. Starting from $\rho^{(0)} = \pi$, each iteration performs the update

$$\rho^{(t+1)} = \Pi_{\Delta_M}\Big(\rho^{(t)} - \eta_t \nabla J(\rho^{(t)})\Big),$$

where Π_{Δ_M} denotes the Euclidean projection onto the simplex, and η_t is chosen by backtracking to ensure sufficient decrease. The gradient admits a closed form:

$$\nabla J(\rho) = \alpha\,\ell + (1-\alpha)\,\frac{1}{n}P^\top(P\rho - Y) + \frac{1}{\beta}\big(\log(\rho/\pi) + \mathbf{1}\big),$$

where the logarithm is taken componentwise. In the implementation, we clip ρ away from zero for numerical stability when evaluating the logarithms. Once the optimizer converges, the aggregated estimator is returned as $\widehat{f}^{\,Q}(x_i) = u_i(\widehat{\rho}) = (P\widehat{\rho})_i$ together with the weights $\widehat{\rho}$.

Why this becomes intractable in high dimension. While the above procedure is practical when the dictionary size M is moderate and all $f_{\theta_j}(x_i)$ can be precomputed and stored, it quickly becomes computationally and memory intensive when M is large (or when θ is continuous). Each gradient evaluation requires at least a matrix-vector product with P and $P^\top$, costing $O(nM)$ operations, and the simplex projection costs $O(M \log M)$ per iteration. Moreover, storing P itself requires $O(nM)$ memory. Therefore, in genuinely high-dimensional aggregation problems (very large dictionaries, continuous parameterizations, or expensive predictors), a direct optimization over $\rho \in \Delta_M$ becomes impractical.

Idea using proposition 5 In opposition to the Exponential Weights estimator, the Q-aggregation one does not admit any explicit Gibbs-form expression. Hence, we can't apply the same strategy to apporximate the Q-aggregation estimator. However, the paper offers a potential avenue for further investigation. Indeed, Proposition 5 shows that the Q-aggregation estimator belongs to a family of distributions that is fully characterized by a finite number of moments, namely the first and second moments of $f_\theta(x_i)$ for each design point x_i, $i = 1, \ldots, n$. As a consequence, computing the Q-aggregation estimator does not require approximating the entire distribution over θ, but only these $2n$ moments.

We then investigated how to use this result. In practice, these moments are unknown and must be approximated. A natural approach consists in introducing a parametric family of distributions $(\rho_\varphi)_{\varphi \in \Phi}$ and approximating the required moments using Monte Carlo samples drawn from ρ_φ. These empirical moments are then plugged into the Q-aggregation objective, which is subsequently minimized with respect to the parameter φ using stochastic gradient methods. However, this approach faces several difficulties. First, the resulting optimization problem is generally non-convex in φ, and therefore stochastic gradient descent does not come with global convergence guarantees. Second, the objective involves nested Monte Carlo approximations, which induces additional stochastic noise. Finally, the quality of the approximation heavily depends on the expressiveness of the chosen parametric family: if ρ_φ is poorly specified, the solution may remain far from the true Q-aggregate. As a result, we think Proposition 5 should primarily be viewed as a structural and theoretical result. Still, it provides closed-form expressions for the Q-aggregate in special cases, such as linear prediction functions combined with Gaussian priors, and suggests that, when the functions are approximately linear and the prior is Gaussian, a Gaussian parametrization of the Q-aggregate may be a reasonable approximation in practice. Beyond these structured settings, however, closed-form solutions do not appear to be available, and the direct practical use of Proposition 5 is limited. As a consequence, the practical computation of the Q-aggregation estimator in general settings does not appear to be fully resolved in the existing literature.

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4 Appendix

4.1 Setting for the computation of Langevin dynamics (ULA) for the closed form case

Data Generation and Statistical Setting We consider a linear regression model with Gaussian noise,

$$Y = X\theta^* + \varepsilon,$$

where $X \in \mathbb{R}^{n \times D}$ is a random design matrix with i.i.d. standard Gaussian entries, $\theta^* \in \mathbb{R}^D$ is the true parameter vector, and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ is a Gaussian noise vector.

In the experiments, the dimension is set to $D = 20$ and the number of observed samples to $n = 1000$. The true parameter θ^* is randomly generated from a centered Gaussian distribution with moderate variance. The response vector Y is obtained by adding Gaussian noise with variance $\sigma^2 = 0.25$.

Empirical Verification of Proposition 2 According to Proposition 2, the exponential weights estimator associated with a quadratic loss and a Gaussian prior coincides with the Ridge regression solution, provided that the parameters β and γ satisfy the prescribed relationship.

In this setting, the exact mean of the Gibbs posterior can be computed in closed form and corresponds to the Ridge estimator $\hat{\theta}_{\text{Ridge}}$. This closed-form solution serves as a reference target for assessing the accuracy of the ULA approximation.

The ULA algorithm is run for a large number of iterations, and the empirical mean of the samples after the burn-in phase is used as an approximation of the posterior expectation.

Convergence Diagnostics for ULA To analyze the convergence of the ULA algorithm, we study the evolution of the ergodic averages of the sampled parameters. The distance between the cumulative mean $\bar{\theta}_t$ and the exact Ridge solution is monitored as a function of the number of post burn-in samples, as shown in Figure 1.

4.2 Setting for the computation of Langevin dynamics (ULA) with Laplace prior

We consider a linear regression model with fixed design given by

$$Y = f^*(X) + \varepsilon = X\theta^* + \varepsilon,$$

where $X \in \mathbb{R}^{n \times D}$ is a fixed design matrix, $\theta^* \in \mathbb{R}^D$ is an unknown parameter vector, and $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ is a Gaussian noise vector.

In our experiments, the dimension is set to $D = 10$ and the sample size to $n = 50$. The true parameter θ^* is chosen to be sparse, with only the first three coordinates being non-zero. The corresponding target function f^* is evaluated on the fixed design points, and the observations Y are generated by adding Gaussian noise with variance $\sigma^2 = 1$.

To estimate the left-hand side of Theorem 1, corresponding to the expected risk of the exponential weights estimator, we repeat the experiment over multiple independent realizations of the noise. For each run, a new dataset is generated according to the same fixed design, and the associated Gibbs posterior is approximated using the algorithm described in section 3.3.

The exponential weights estimator is then obtained by averaging the ULA samples, and its risk $\|f_{\hat{\theta}_{EW}} - f^*\|_n^2$ is computed. Averaging this quantity over all repetitions provides an empirical estimate of the expectation appearing on the left-hand side of the theorem, which is subsequently compared to the theoretical upper bound.

To assess the convergence of the ULA algorithm, we run several independent chains initialized with different random seeds and plot the firt parameter at each time after burn-in.

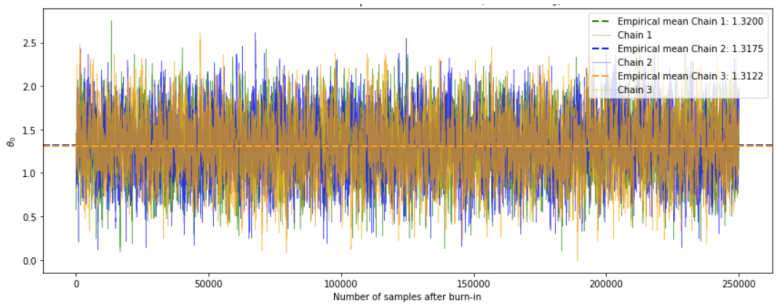


Figure 3: Convergence of the first parameter for three different chains generated via ULA

Then, for each parameter coordinate, we compute the Gelman–Rubin diagnostic $\hat{R}$.

The within-chain and between-chain variances are combined to form the potential scale reduction factor, and convergence is declared when $\hat{R} < 1.05$.

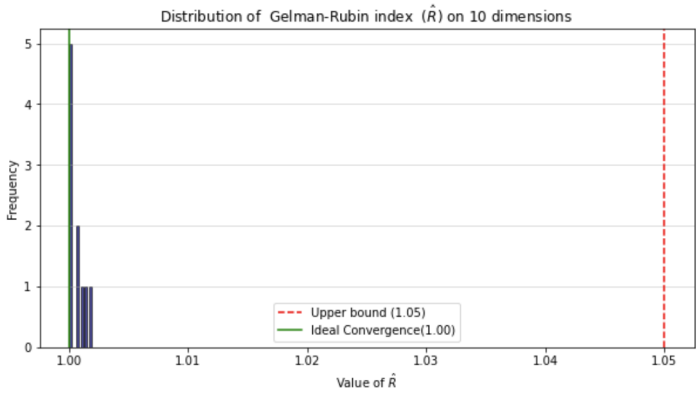


Figure 4: Histogram of the Gelman-Rubin index for each parameter

This diagnostic provides empirical evidence that the SGLD sampler has reached a stable stationary regime and that the resulting samples can be used reliably for estimation and inference.